- MYALGIC ENCEPHALOMYELITIS

A CMPS 262 Project by: Nadjib Tebbal & Fadel Mansour

Made with GAMMA

Multi-Layer Data Science Investigation of ME/CFS

Analyzing the Complexities of Myalgic Encephalomyelitis/Chronic Fatigue Syndrome

Made with GAMMA

The Challenge of ME/CFS Heterogeneity

Myalgic Encephalomyelitis / Chronic Fatigue Syndrome (ME/CFS) is a complex and often debilitating condition whose pathophysiology remains poorly understood, making precise diagnosis challenging.

Defining Symptoms

- Profound, non-restorative fatigue
- Post-Exertional Malaise (PEM)
- Cognitive dysfunction ("brain fog")

Diagnostic Overlap

- Frequent overlap with clinical depression
- Mimics symptoms of hypothyroidism
- Need for robust biological differentiators

Made with GAMMA

A Multi-Layered Data Science Approach

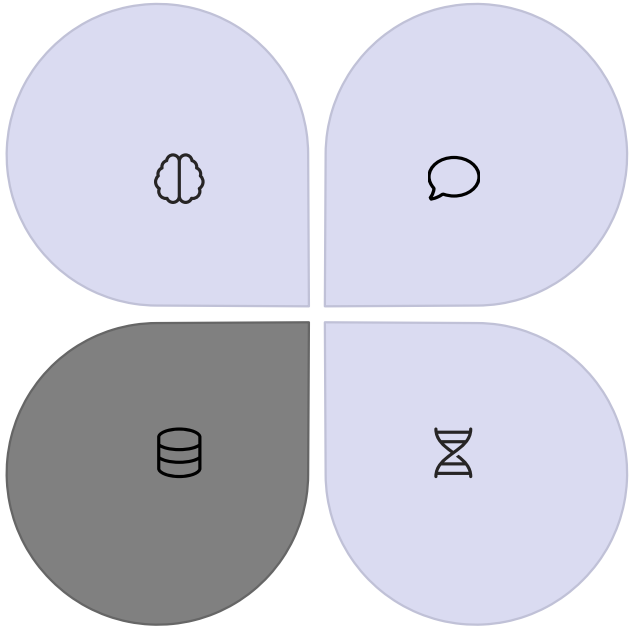
Our investigation is structured across three distinct biological and clinical levels to capture the systemic nature of ME/CFS, moving from high-level symptoms to molecular mechanisms.

1. Symptom-Level

Identifying differentiating clinical patterns and patient phenotypes using NLP on medical texts.

2. Metabolic-Level

Exploring altered biochemical pathways and biomarkers through comprehensive metabolomics data analysis.



3. Genetic/Epigenetic-Level

Analyzing differences in gene expression and DNA methylation patterns (Omics data).

4. Cross-Layer Integration

Comparing consistent findings across all layers for robust signature discovery.

Made with GAMMA

Diverse Datasets for Comprehensive Analysis

We leverage a spectrum of data types, from unstructured clinical notes to highly structured molecular measurements, ensuring broad coverage of the disease landscape.

Unstructured Text

Synthetic medical text data (Kaggle) for NLP-based symptom extraction.

Metabolomics

Comprehensive datasets detailing small-molecule profiles.

1

2

3

4

Clinical Cohorts

Tabular datasets comparing ME/CFS and depression groups; Longitudinal follow-up data (Bristol NHS).

Omics Data

NCBI GEO: GSE251872 (Gene Expression) and GSE111183 (DNA Methylation).

Made with GAMMA

Symptomatic approach (Layer 1) : Part 1

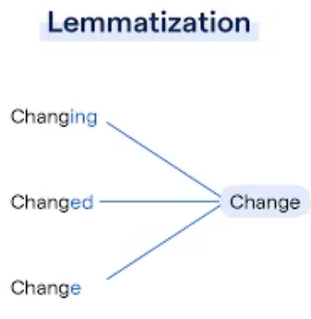
Characteristics of Myalgic Encephalomyelitis:

DATASET 1: Medical data gathered about patient complaints, professional diagnosis , and perscribed treatment

Contains ~400 instances, with around 2.7% of the cases being Chronic fatigue syndrome

Key NLP Methods

- **TF-IDF (Term Frequency-Inverse Document Frequency):** Identifying term importance.
- **Keyphrase Extraction:** Pinpointing critical, defining symptoms.
- **Clustering (e.g., K-Means):** Discovering distinct patient symptom patterns (phenotypes).
- Taking the difference of mean vectors to get the ME/CFS related words from the dataset



$$TF(t,d) = \frac{number\ of\ times\ t\ appears\ in\ d}{total\ number\ of\ terms\ in\ d}$$

$$IDF(t) = log\frac{N}{1+df}$$

$$TF-IDF(t,d) = TF(t,d) * IDF(t)$$

Top ME/CFS-related words:
- tired
- sleep
- full
- feeling
- even
- lately
- fatigue
- extreme
- constant
- pain
- minimal
- lethargic

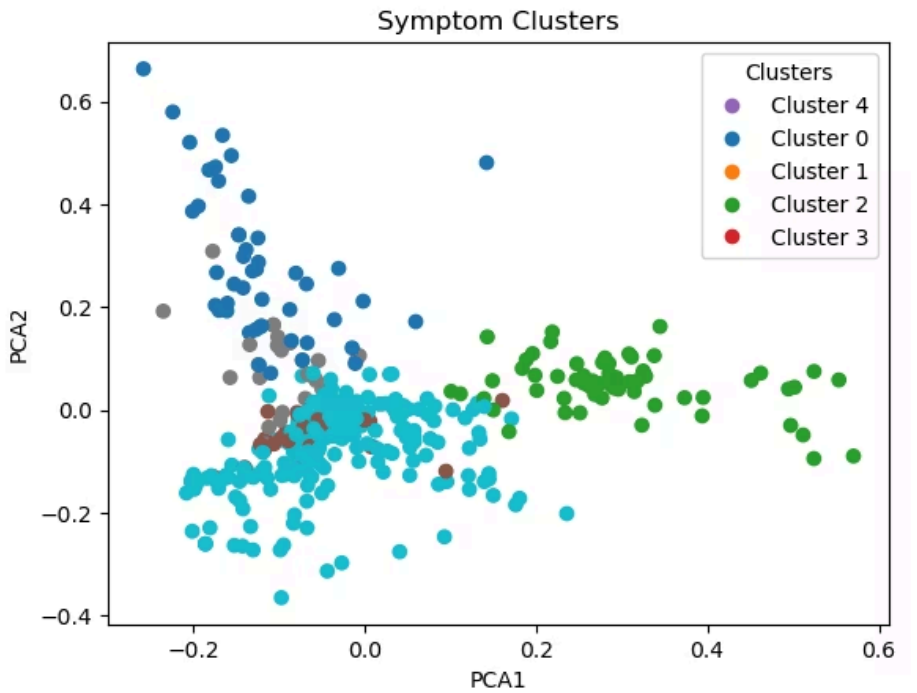
Made with GAMMA

Symptomatic approach (Layer 1) : Part 1 - Follow up

KNN Algorithm with N = 5

Generated 5 clusters based on patient symptoms, putting ME/CFS in the same category as the Diabetes family, Vagal Synocope and others!

PCA with 2 components:



This doesn't reveal too much info so further EDA is needed

Made with GAMMA

Symptomatic approach (Layer 1) : Part 1 - Follow up

Let's look at diseases with same patient description using NLP (COSINE SIMILARITY):

Some Findings:

1. Muscular dystrophy and hepatitis share superficial symptoms with ME but differ clinically, excluding them avoids false positives.
2. Sleep apnea, insomnia, iron deficiency, and hypothyroidism recur and may explain some ME misclassifications.
3. This suggests examining **sleep patterns, thyroid, and iron levels** in ME patients.
4. Depression and anxiety frequently co-occur. This is the main focus of exploration in Dataset 2.

```
vectorizer = TfidfVectorizer(stop_words='english') X =
vectorizer.fit_transform(df['Patient_Problem'])

Identify ME/CFS rows

mecfs_mask = df['Disease'].str.contains('fatigue|cfs', case=False,
na=False)

Create mask for "other diseases" only

other_mask = ~mecfs_mask

Cosine similarity between ME/CFS rows and all rows

similarities = cosine_similarity(X[mecfs_mask], X[other_mask])

Get indices of other rows (non-ME/CFS)

other_indices = np.where(other_mask)[0]

For each ME/CFS row, find the top 5 most similar complaints from
other diseases.
```

Made with GAMMA

Symptomatic approach (Layer 1) : Part 1 - Follow up

- Shortlisting the most frequent similar conditions

Condition	Count
ME/CFS	11
Insomnia	11
Hypothyroidism	13
Iron Deficiency Anemia	2
Depression	5
Anxiety Disorder	4
Sleep Apnea	11

Observation: Similar frequencies across ME/CFS and fatigue-related disorders suggest possible overlap or diagnostic confusion.

Made with GAMMA

Symptomatic approach (Layer 1) : Part 1 - Key insights

Let's look at among the MECFS complaints, which ones are shared in the other diseases

```
def get_top_keywords(df, disease_name, col='Patient_Problem', top_n=10):
# Select rows for this disease
text_data = df[df['Disease'].str.contains(disease_name, case=False, na=False)][col].tolist()
if not text_data:
return []

# Get top N keywords
top_indices = scores.argsort()[::-1][:top_n]
top_keywords = [terms[i] for i in top_indices]
return top_keywords


conditions = ['Insomnia', 'Hypothyroidism', 'Iron Deficiency Anemia', 'Depression', 'Major depressive disorder', 'anxiety disorder',
'sleep apnea']

condition_keywords = {cond: get_top_keywords(df, cond) for cond in conditions}

mecfs_mask = df['Disease'].str.contains('fatigue|cfs', case=False, na=False) mecfs_df = df[mecfs_mask]

for cond, kws in condition_keywords.items(): print(f"\n--- ME/CFS complaints potentially matching {cond} ---") for idx, row in
mecfs_df.iterrows(): if any(kw.lower() in row['Patient_Problem'].lower() for kw in kws): print(f"- {row['Patient_Problem']}")
```

Interpretation:

- Sleep, thyroid, and iron issues are **likely comorbid** rather than distinct.
- These are **treatable**, unlike ME/CFS misclassification may occur when underlying causes aren’t fully investigated.
- Focus should shift to distinguishing **ME/CFS from mood and psychosomatic disorders** (e.g., depression, anxiety) rather than purely physiological ones.

Made with GAMMA

Symptomatic approach (Layer 1) : PART 2
ME CFS vs DEPRESSION DATASET

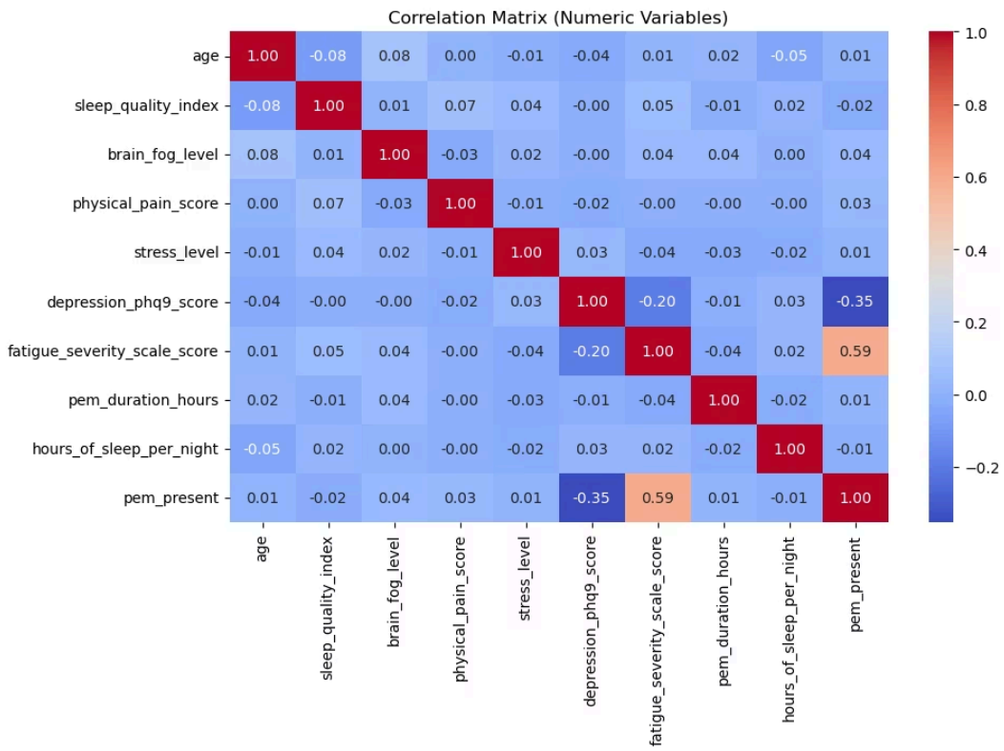
Made with GAMMA

Symptomatic approach (Layer 1) : Part 2
ME CFS or Depression?

DATASET 2: Synthetic dataset with 1001 entries about cases of ME/CFS, depression, and both at the same time

Missing values :

< 5% missing values overall, however since this is a medical dataset we can't just simply impute



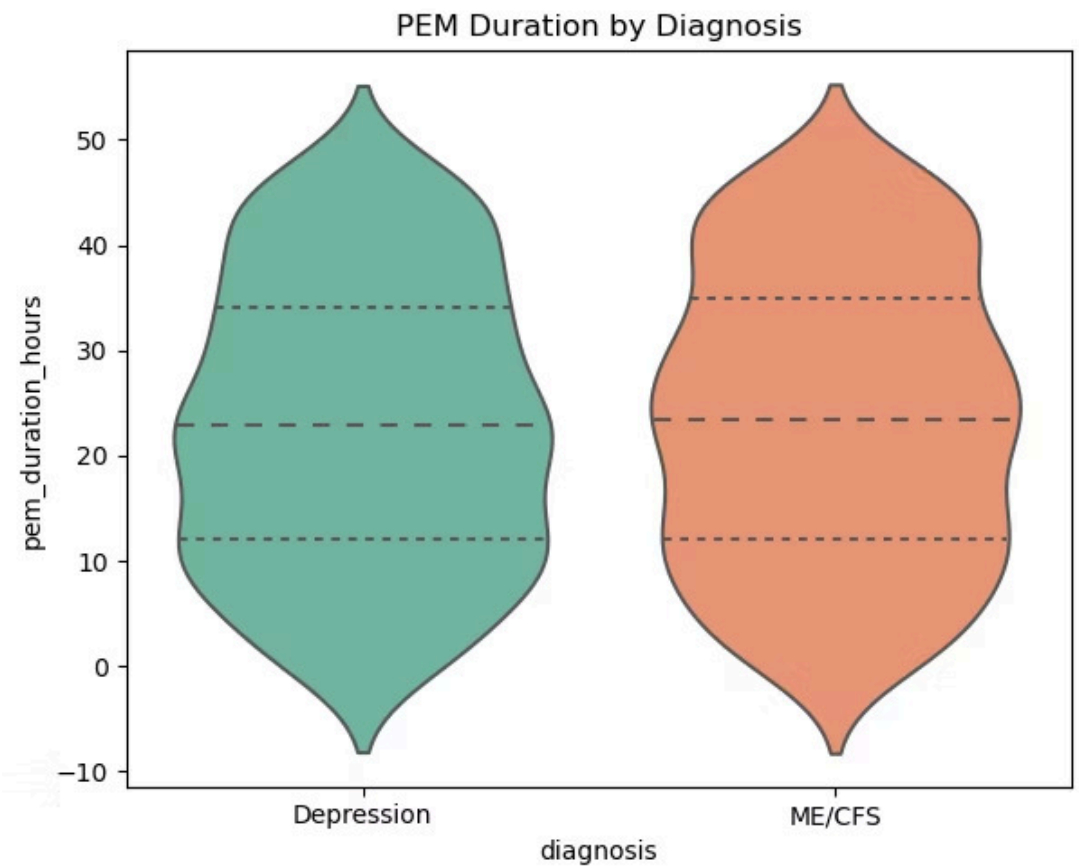
- Strong correlation between fatigue levels and peak PEM hours 60%
- Some correlation between depression scores and PEM/fatigue 39% and 20% respectively

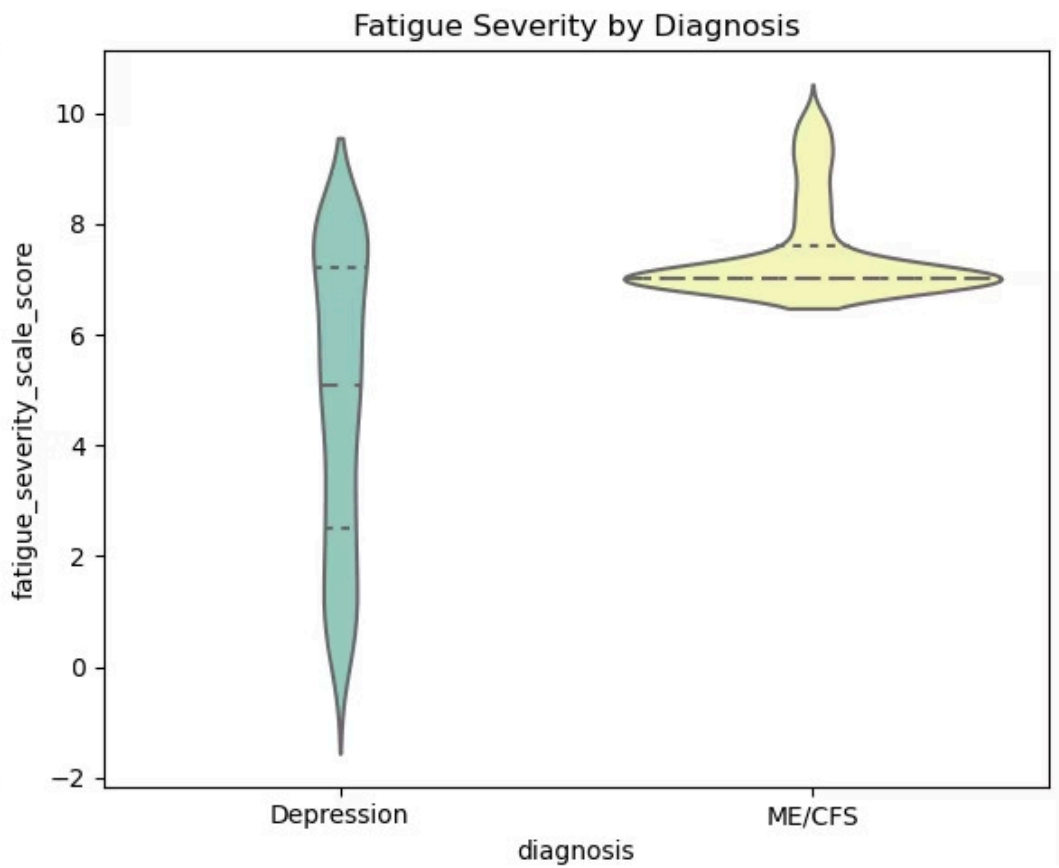
Imputation Strategy Summary

1. **Targeted (1D) Imputation**
   - Filled missing values in **fatigue severity** and **PEM duration** using the **mean within each diagnosis group** (ME/CFS vs Depression).
2. **Contextual (2D) Imputation**
   - For **depression scores**, used **mean within both diagnosis and fatigue quartile groups**.
3. **Remaining cols - > Mean Imputation**

Made with GAMMA

Symptomatic approach (Layer 1) : Part 2
Some interesting EDA and insights:

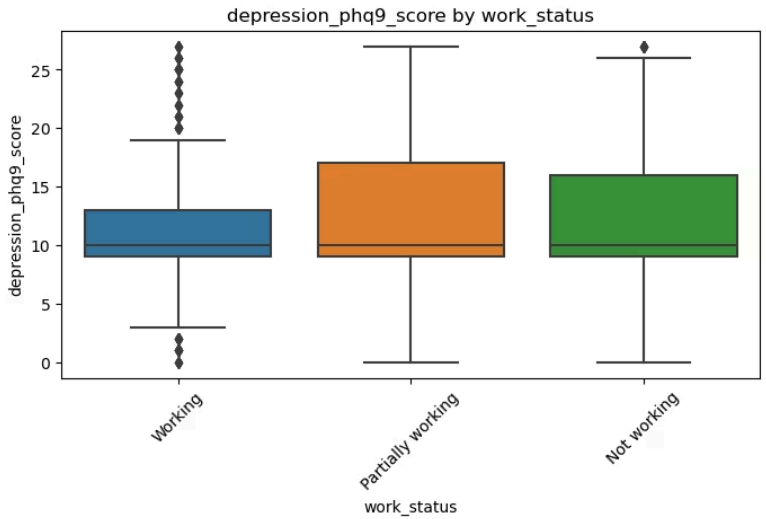




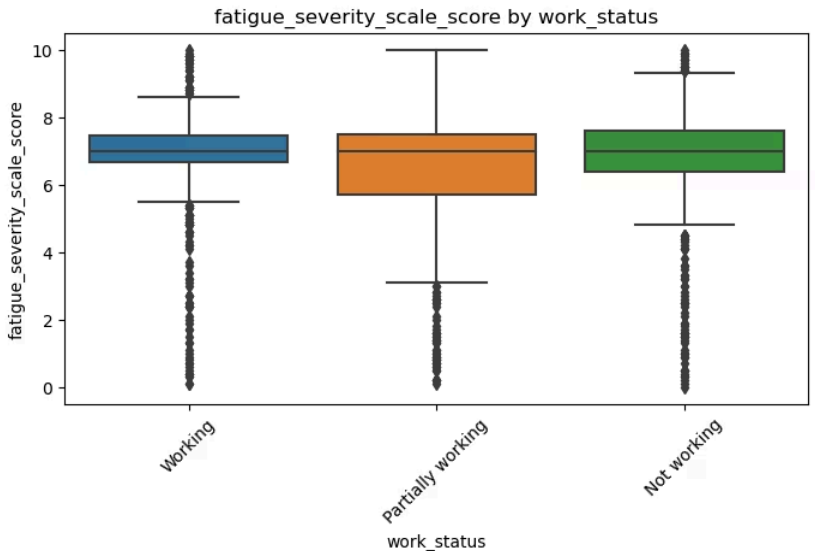
- The PEM variable though usually crucial is mostly noisy and inconsequencial in this dataset
- ME/CFS patients report much much higher levels of fatigue with the mean being around 7
- Most other variables follow an approximately normal distribution

Made with GAMMA

Symptomatic approach (Layer 1) : Part 2
Some interesting EDA and insights:



- People who work are more likey to either be happy or very sad



- Similarly they're unpredictabely tired compared to people who work part time
- Though for the unemployed patients, they report more extreme values as well, this indicates strong disability

Made with GAMMA

Symptomatic approach (Layer 1) : Part 2
How the conditions affect work

Diagnosis	Working (%)	Partially Working (%)	Not Working (%)
ME/CFS	**40.3**	33.2	26.4
Depression	**29.7**	39.9	30.4
Both	**28.2**	31.7	**40.1**

Key Insights

- **ME/CFS patients** show higher full-time work participation (40%) than those with depression (30%).
- **Depression** is more disabling overall, reducing full-time employment and increasing partial work.
- **Co-occurrence (ME + Depression)** produces the **lowest work engagement**, suggesting compounded functional impairment.

Made with GAMMA

Quick Survey:

Men VS Women

Do you believe men on average cope better with depression than men?

Made with GAMMA

Symptomatic approach (Layer 1) : Part 2
How the conditions affect work

Diagnosis	Gender	Not Working (%)	Partially Working (%)	Working (%)
Depression	Female	38.6	30.7	30.7
	Male	**41.6**	32.7	**25.7**
ME/CFS	Female	**22.6**	**36.6**	**40.9**
	Male	29.9	30.3	39.8
Both	Female	22.6	36.6	40.9
	Male	29.9	30.3	39.8

Key Insights

- In **Depression**, women are **slightly more likely to work** than men (30.7% vs 25.7%), while men show higher full unemployment (41.6%).
- In **ME/CFS**, both genders work at similar rates, but **females have fewer not working** (22.6% vs 29.9%).
- When **both conditions coexist**, the gender gap persists , **men appear more disabled** by ~7–8%, confirming your observation.

Made with GAMMA

Symptomatic approach (Layer 1) : Part 2
How the conditions affect work

Diagnosis	Gender	Not Working (%)	Partially Working (%)	Working (%)
Depression	Female	38.6	30.7	30.7
	Male	**41.6**	32.7	**25.7**
ME/CFS	Female	**22.6**	**36.6**	**40.9**
	Male	29.9	30.3	39.8
Both	Female	22.6	36.6	40.9
	Male	29.9	30.3	39.8

Key Insights

- In **Depression**, women are **slightly more likely to work** than men (30.7% vs 25.7%), while men show higher full unemployment (41.6%).
- In **ME/CFS**, both genders work at similar rates, but **females have fewer not working** (22.6% vs 29.9%).
- When **both conditions coexist**, the gender gap persists , **men appear more disabled** by ~7–8%, confirming your observation.

Made with GAMMA

Symptomatic approach (Layer 1) : Part 2
How the conditions affect work

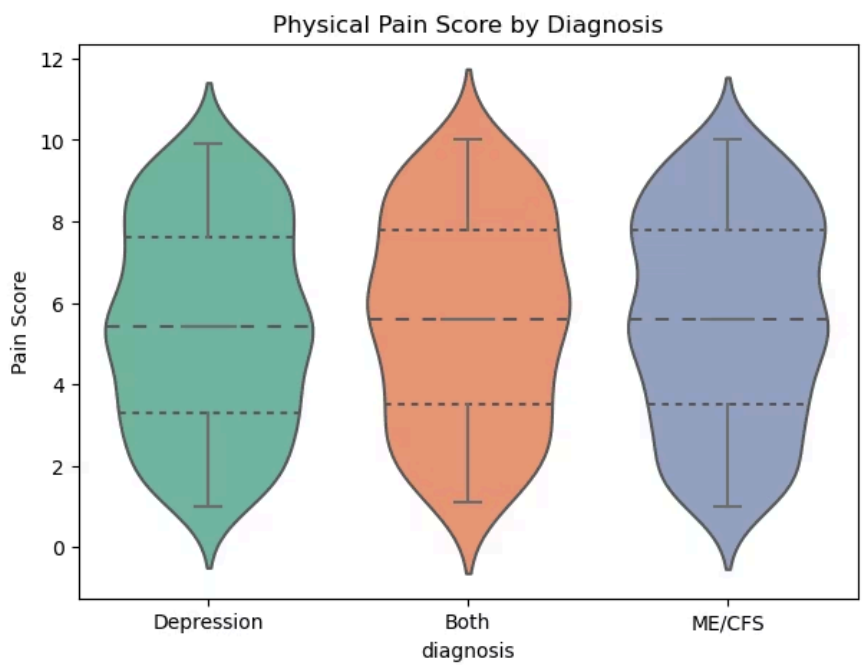
Diagnosis	Gender	Not Working (%)	Partially Working (%)	Working (%)
Depression	Female	38.6	30.7	30.7
	Male	**41.6**	32.7	**25.7**
ME/CFS	Female	**22.6**	**36.6**	**40.9**
	Male	29.9	30.3	39.8
Both	Female	22.6	36.6	40.9
	Male	29.9	30.3	39.8

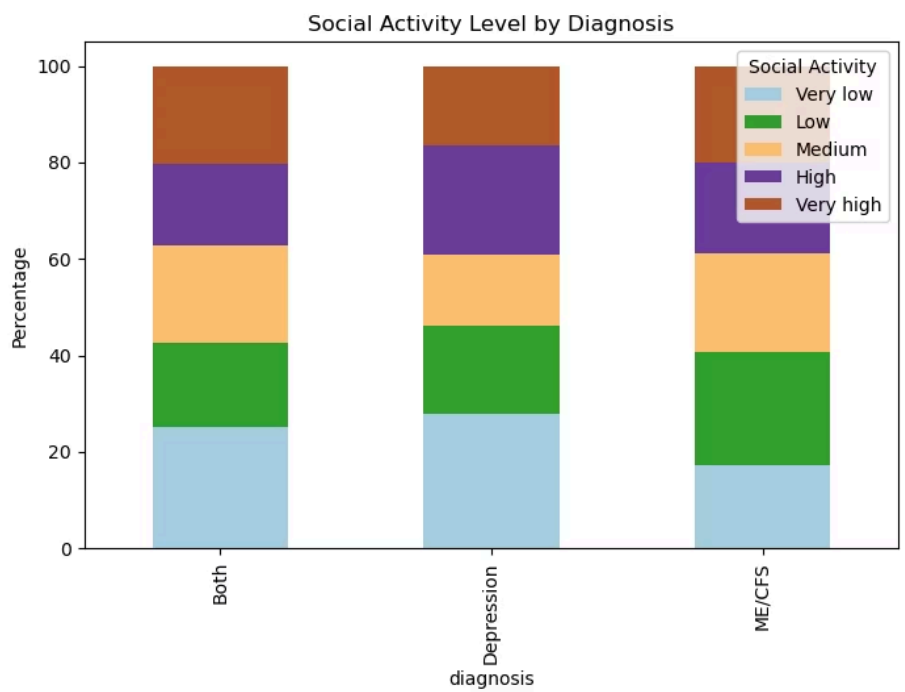
Key Insights

- In **Depression**, women are **slightly more likely to work** than men (30.7% vs 25.7%), while men show higher full unemployment (41.6%).
- In **ME/CFS**, both genders work at similar rates, but **females have fewer not working** (22.6% vs 29.9%).
- When **both conditions coexist**, the gender gap persists , **men appear more disabled** by ~7–8%, confirming your observation.

Made with GAMMA

Symptomatic approach (Layer 1) : Part 2
Final insights





- Similar reported pain levels, though the **standard deviation** is huge (2.5) so this doesnt tell the full story

- **ME/CFS:** Social activity is **consistently limited**, mostly in **Low–Medium** categories.
- **Depression:** Social activity is **more variable**, with higher proportions in **Very Low** and **Very High** categories.

Made with GAMMA

Next Milestone:

METABOLOMICS and 12-Month/Long-term follow up

Real World NHS dataset from the University of Bristol

Made with GAMMA

Molecular Mechanics: Metabolomics and Omics

Layers 2 and 3 focus on deep molecular analysis to uncover underlying biological perturbations in ME/CFS patients.

Metabolite Analysis

Using **PCA (Principal Component Analysis)** and **ANOVA** to identify significantly altered metabolites and metabolic pathways compared to controls.

Gene Expression

Differential expression analysis using tools like GEO2R to find up/down-regulated genes. Visualization via **Volcano Plots**.

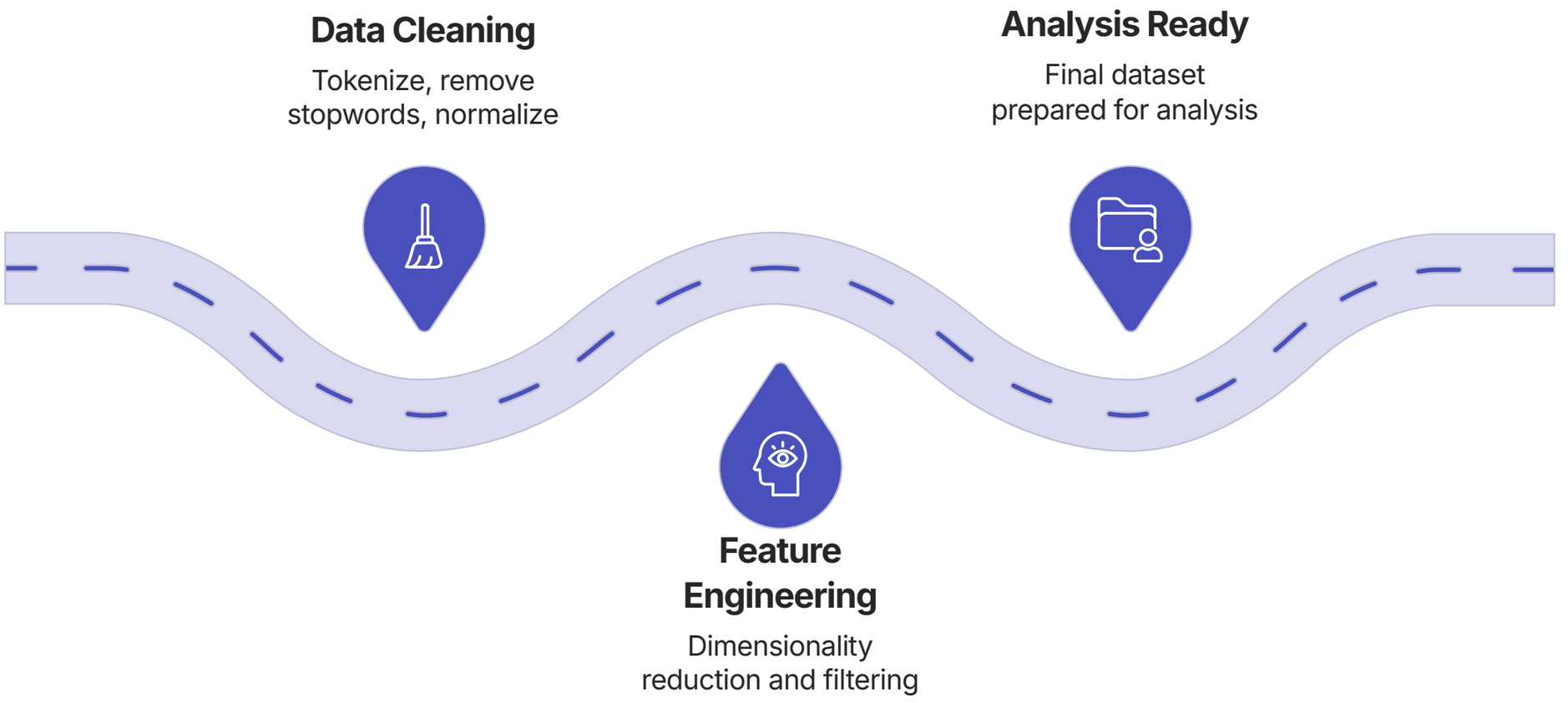
DNA Methylation

Exploring epigenetic differences through differential methylation analysis, aiming to explain altered gene function.

Made with GAMMA

Data Preprocessing Pipeline

Rigorous data cleaning and feature engineering are essential steps for deriving reliable insights from complex biological data.



1

Cleaning

Tokenization, stopword removal, normalization, and low-variance filtering.

2

Feature Engineering

Dimensionality reduction (PCA/t-SNE) and significance filtering ($p < 0.05$).

3

Feature Ranking

Using feature importance scores from ML models to prioritize key biological markers.

Made with GAMMA

Planned Analytical Toolkit

A combination of advanced machine learning and statistical methods will be employed to maximize data yield and ensure robust conclusions.

Clustering

Unsupervised learning for phenotype stratification.

Regression

Modeling relationships between clinical variables and molecular data.

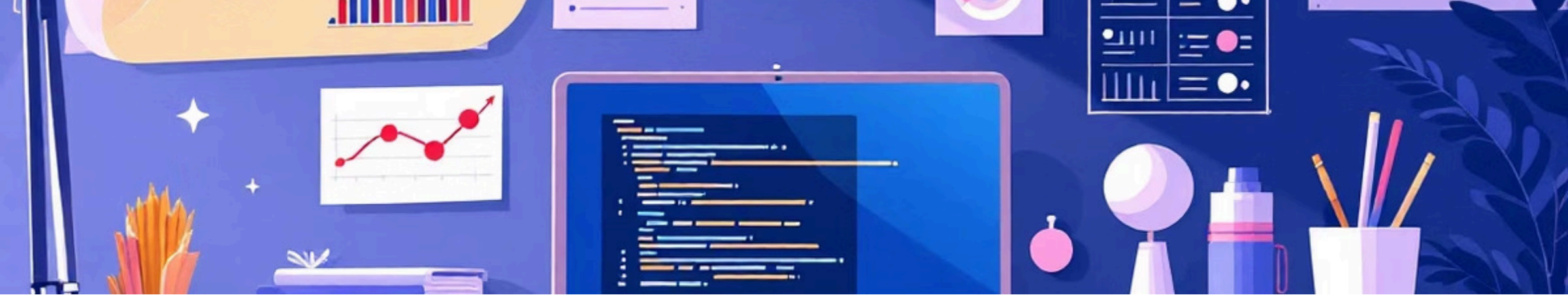
Random Forest

Classification for diagnostic differentiation and feature importance.

ANOVA

Identifying statistically significant differences across groups (e.g., control vs. patient).

Made with GAMMA



Execution Strategy and Tools

The bulk of computational analysis will utilize the Python ecosystem, supplemented by specialized bioinformatics tools for omics data processing.

Python Ecosystem

Used for NLP, Classification Modeling (Scikit-learn), Feature Engineering, and Metabolomics analysis.

GEO2R (NCBI)

Dedicated platform for performing differential expression and methylation analysis on NCBI GEO datasets.

Made with GAMMA

Expected Impact and Future Direction

Successful cross-layer integration is anticipated to yield robust, multimodal signatures that can inform better clinical practice and future research.

Improved Diagnostic Models

Consistent signatures across layers provide a strong basis for new, objective ME/CFS diagnostic criteria.

Validated Biomarkers

Identification of specific metabolic, genetic, or symptomatic features for targeted therapeutic development.

Hypothesis Generation

The work will generate new, testable hypotheses regarding ME/CFS etiology and pathophysiology.

Made with GAMMA